

Pre-Class System Setup Instructions for DSML

Before diving into Machine Learning, let's set up our system to ensure a smooth workflow for all assignments and projects.

Step 1: Install Anaconda

Choose the appropriate link based on your system configuration:

- **Windows (64-bit):** [Download](#)
- **Mac:** [Download](#)
- **Linux (64-bit):** [Download](#)

Installation Steps:

1. Download the installer for your system.
2. Run the installer and ensure the installation path does not contain spaces.
3. Check the option **"Add Anaconda to PATH"** (this is crucial to avoid issues while installing packages).
4. Complete the installation and verify it by opening Anaconda Prompt.
5. Installing VS Code is optional—you can skip it if you prefer.

Step 2: Install Required Packages

We'll install and verify the following packages:

1. **Plotly** – For interactive visualizations and dashboards.
2. **Gensim** – For topic modeling and word embeddings (NLP).
3. **Graphviz** – For visualizing decision trees, Dask dataframes, and deep learning models.
4. **Distance** – For computing text similarity (installed via pip).
5. **FuzzyWuzzy** – For fuzzy string matching and text similarity.
6. **Imbalanced-learn (imblearn)** – For handling imbalanced datasets in machine learning.
7. **NetworkX** – For analyzing and visualizing complex networks/graphs.
8. **spaCy** – For natural language processing (NLP).
9. **Foulm** – (Check package purpose; possibly a typo for "Folium").
10. **TensorFlow (CPU only)** – Essential for deep learning and neural networks.
11. **Keras** – High-level API for building deep learning models.
12. **PrettyTable** – For creating formatted tables in Python.
13. **TQDM** – For showing progress bars in loops.
14. **XGBoost** – For gradient boosting, a powerful machine learning algorithm.
15. **GPXPY** – For parsing and manipulating GPS data.

16. **NLTK Stopwords** – For text preprocessing in NLP tasks.
17. **Scikit-multilearn** – For multi-label classification.
18. **Surprise** – For building recommendation systems.
19. **StellarGraph** – For graph-based machine learning.
20. **TensorBoard** – For visualizing deep learning model training.

Step 3: Install Required Packages

Now that Anaconda is installed, we need to install essential libraries for machine learning. These packages will help with data processing, visualization, and model building.

Open Anaconda Prompt (Windows) or Terminal (Mac/Linux)

To install packages, first open:

- **Windows:** Search for "**Anaconda Prompt**" and open it.
- **Mac/Linux:** Open **Terminal**.

Install Packages using Conda:

Run the following command to install multiple packages at once:

```
conda install -c conda-forge plotly gensim graphviz imbalanced-learn  
networkx spacy tensorflow keras prettytable tqdm xgboost gpxpy nltk  
scikit-multilearn surprise stellargraph tensorboard
```

Step 5: Verify Installation

To make sure everything is installed, open **Jupyter Notebook**:

1. Restart Jupyter Notebook to ensure all packages are properly installed.
2. Check package functionality by importing them in Python. Copy and run the following code inside a code cell:

```
import plotly, gensim, graphviz, imblearn, networkx, spacy, tensorflow,  
keras  
import prettytable, tqdm, xgboost, gpxpy, nltk, skmultilearn, surprise,  
stellargraph, tensorboard  
import distance, fuzzywuzzy
```

```
print("All packages installed successfully!")
```

If everything is installed correctly, you should see:

```
All packages installed successfully!
```

Now, your system is fully set up and ready for Machine Learning! 🚀